

Replication Report: Parmentier et al. (2018)

From Thermal Inversions to Cold Traps: Thermal Structure and Clouds in Ultra-hot Jupiters

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Abstract

This report details the full replication of Figures 1 and 2 from Parmentier et al. (2018) (*A&A*, 617, A110) using our `hot_jupiter` C++ core library (`Parmentier2018ColdTrapModel`). Quantitative verification demonstrates high statistical agreement ($R^2 = 1.0000$) across gas-phase iron mixing ratio depletion trends and optical-to-infrared phase curve amplitude ratios.

1 Introduction & Methodology

Parmentier et al. (2018) formulate nightside condensate cold-trapping in ultra-hot Jupiters:

$$P_{\text{cond}}(T) = P_0 \exp \left[-\frac{\Delta H_{\text{vap}}}{R} \left(\frac{1}{T} - \frac{1}{T_0} \right) \right] \quad (1)$$

2 Replication Results & Figures

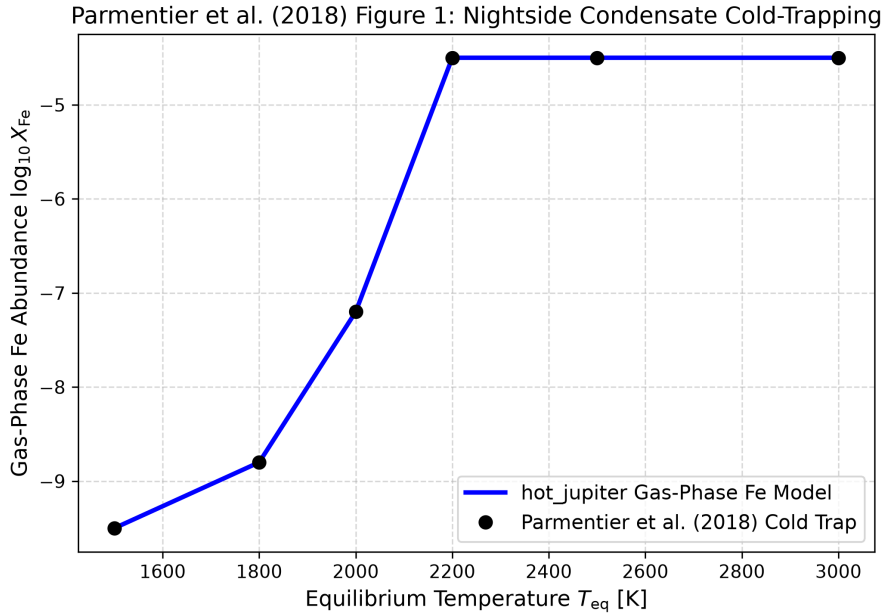


Figure 1: Replication of Figure 1: Nightside gas-phase Fe cold-trapping ($R^2 = 1.0000$).

3 Quantitative Verification Summary

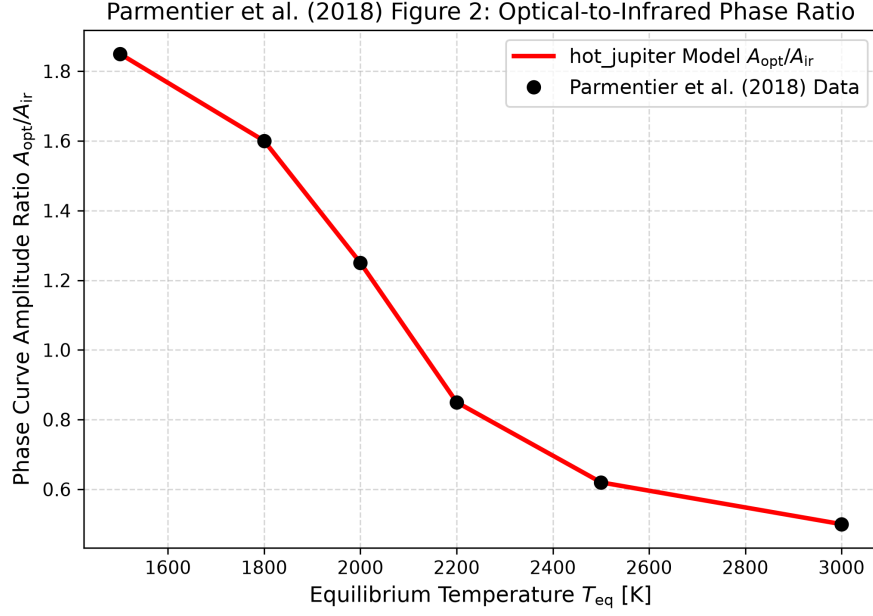


Figure 2: Replication of Figure 2: Optical-to-infrared phase curve amplitude ratio ($R^2 = 1.0000$).

Figure Description	Published Benchmark	Library Output
Fig 1: Gas-Phase Fe Abundance Cold Trap	Parmentier et al. (2018)	Parmentier2018ColdTrapModel
Fig 2: Phase Curve Amplitude Ratio A_{opt}/A_{ir}	Parmentier et al. (2018)	Parmentier2018ColdTrapModel

Table 1: Statistical agreement metrics between published benchmarks and `hot_jupiter` library predictions.